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Task: Peer Review for Module 3

Title: Tube Arcs in X-ray systems

Course: CAS in Applied Data Science

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University: University of Bern

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Summary of Project

My colleague, Sandro Weidmer, completed a Module 3 project on predicting the number of tube arcs per generator device. My colleague works at Comet, where they specialize in plasma control and X-ray technologies. They provide components and systems for manufacturing, communication, and mobility sectors.

The goal of this project was to estimate the relation between the number of tube arcs per generator device and its features. The X-ray generator writes diagnostic reports in a format, and the dataset is obtained from this. It's a regression project where you have both discrete and continuous features.

Key features included:

- Exposure time per device
- Operating point statistics (HV and emission current: max, mean, std)
- Temperature statistics (max, mean, std)
- Device type (Quantitative vs categorical)

The prediction target variable is the: Number of arcs per X-ray generator device.

Strengths and Good Solutions

Good solutions were implemented for this project, as it highlights the steps taken to conduct a data science project. The notebook was well organized, with clear explanations and comments for each code block. This made the analytical process easy to follow and demonstrated a good understanding of each step.

The strengths highlighted in this project are:

The data was sourced and prepared. A PySpark DataFrame was used to process a large dataset, and databricks ensured that it ran smoothly. The filtering based on Device ID and Serial Number ensured that we were only looking at the Device IDs and serial numbers of the arcs sold to customers. This is important, as we only want to look at the relevant data for this project.

The preprocessing phase showed the total number of arcs per generator device. There is a table that shows the total number of arc devices and how many are field devices, and how many are lab devices. This is good to show, as it can help us to understand the distribution between field and lab devices. Histograms were used to show the frequency of the total arcs per device and the exposure time for the devices. Both histograms were skewed, indicating that the data is not normally distributed and is asymmetric. Skewed histograms can lead to poor accuracy and inaccurate predictions for machine learning.

The Scatter plot graph was well visualized, as it showed us the relationship between the arcs and the total exposure time. The results showed that devices with higher arc counts have higher accumulated exposure hours, indicating an increase in usage time and arc occurrence. By grouping the data by tube type, the visualization showed easy comparison of performance across different device categories, helping to identify which tube types showed stable behavior.

Linear regression was used as a baseline model for Machine Learning. We will have to start with something simple to see if it works for us. If the performance is bad, then we should use Random Forest, Gradient Boosting, or XGBoost to improve the performance. In this case, all three models performed poorly, so a feature importance analysis was conducted to identify which features best fit the models.

Three variables had a significant score in the feature importance analysis, and based on this, removing the outliers was a good solution to fit the dataset into the model. The outliers were chosen based on the feature importance analysis. XGBoost was used for the outliers, and it resulted in higher performance.

Finally, the project addressed missing values by removing incomplete records, ensuring cleaner and more reliable data for model training.

Suggestions for Improvements

The results from the Linear Regression, Random Forest, and XGBoost models suggest that the dataset is overfitting. Several approaches could help address this issue.

Cross-validation could be introduced to better evaluate model performance and reduce the risk of overfitting. This can be used to measure the model's quality and accuracy before fitting the dataset into the model. This can help to measure which model gives the best accuracy, and it can help with the fine-tuning of the models.

Using a train, validation, and test split would also help build a more reliable model by ensuring that patterns are learned and evaluated properly on separate subsets of the data. It helps with the generalization across the different dataset partitions.

Tools available in Databricks, such as AutoML and Feature Store, could streamline many aspects of the modeling workflow. AutoML can automatically compare multiple algorithms and hyperparameter configurations, while the Feature Store allows for consistent feature engineering and can be reused across experiments.

Another improvement that could be considered is tracking the poor MAE and R2 results and try to use hyperparameters to fine-tune the model performance. This can improve the accuracy of the models used.

By applying these improvements, the models are more likely to generalize well and produce more reliable results.

Conclusion

Overall, the project is well structured, with the main objective being highlighted and the factors influencing the number of tube arcs per generator device being explored. This project demonstrates a solid understanding of a data science project. From SQL extraction to preprocessing, analysis, model, and evaluation, it provides statistical insights into how exposure time, operating conditions, and device characteristics relate to arc occurrence. While the machine learning models did not achieve strong predictive performance, the project successfully identified key challenges such as overfitting and feature limitations.

The suggestions for cross-validation, training validation test split, and the integration of databricks tools present practical next steps that could significantly strengthen the flow of the project. In summary, this project forms a strong foundation for continued investigation into arc prediction and highlights valuable opportunities for improving model reliability and operational understanding.

References:

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